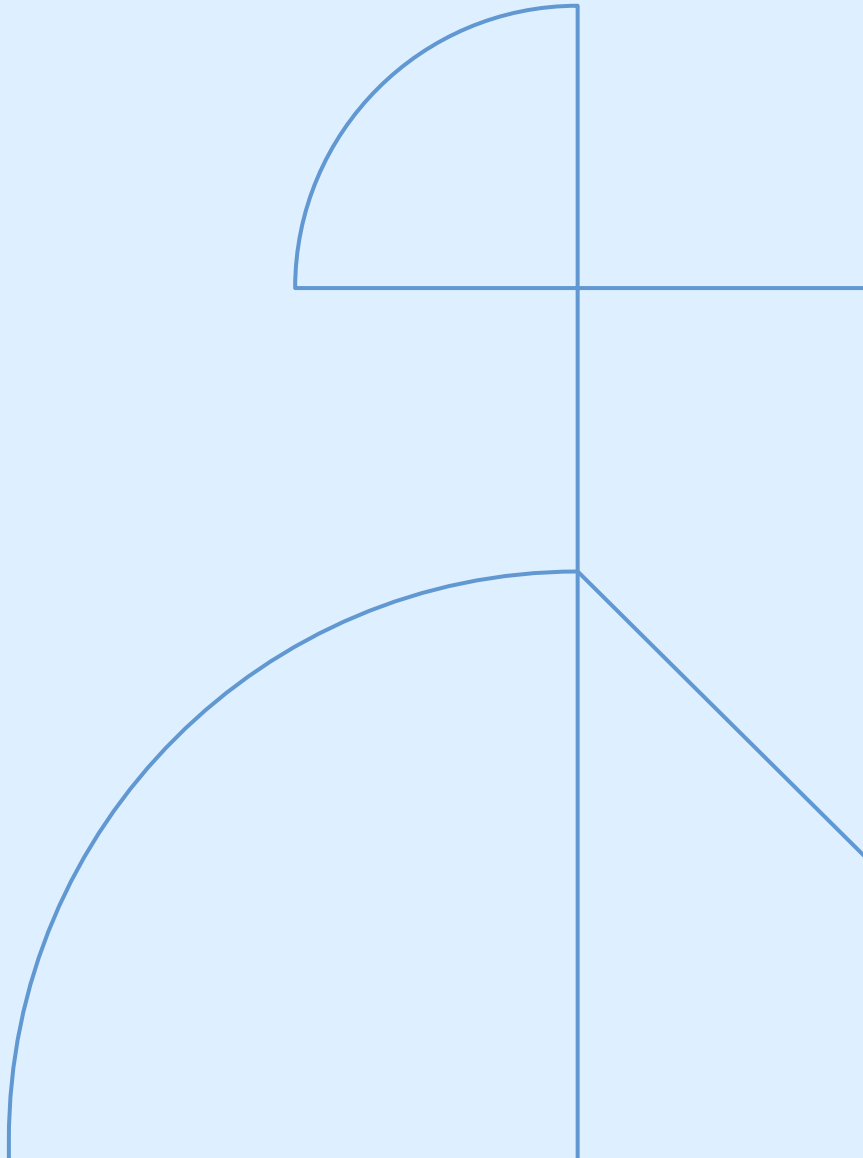


Discrete Event Simulation

Why we simulate?

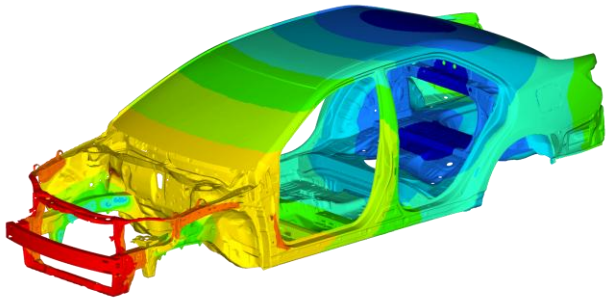


Best Model

- A model can be defined as a simplified representation of some aspects of a real system

- Simplification of the model facilitates easier analysis and interpretation of the system's behavior and outcomes
- Overcomplicated models are useless
- Focusing on specific aspects allows for targeted exploration and understanding of key dynamics or phenomena within the system, without unnecessary complexity
- A model based on a real system ensures relevance and applicability, providing insights and predictions that can inform decision-making and problem-solving in real-world scenarios



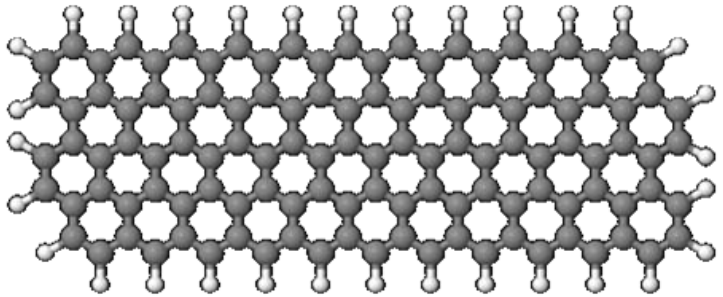


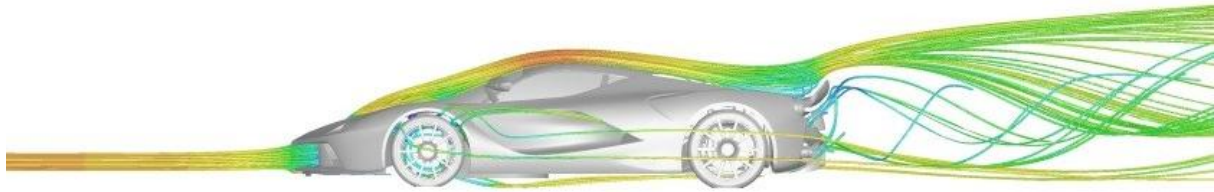


speed=distance/time



CAR





Discrete Event Simulations (DES)

Some Theory

SYSTEM

A collection of entities that act together toward the accomplishment of some logical goal

DYNAMIC SYSTEM

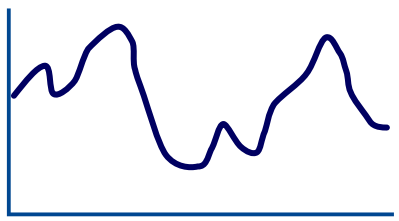
System states change with time

RESOURCES

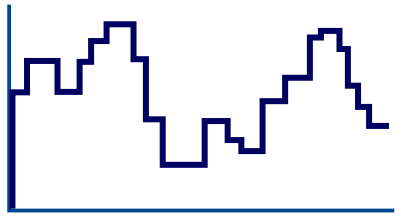
Part of system that is needed by the simulated system to perform certain tasks or activities

Entities:
- People
- Parts
- Commodities
- Messages
- etc.

Actions or events:
- arrival
- departure
- start/stop of process
- overflow
- etc.



Continuous Dynamic System

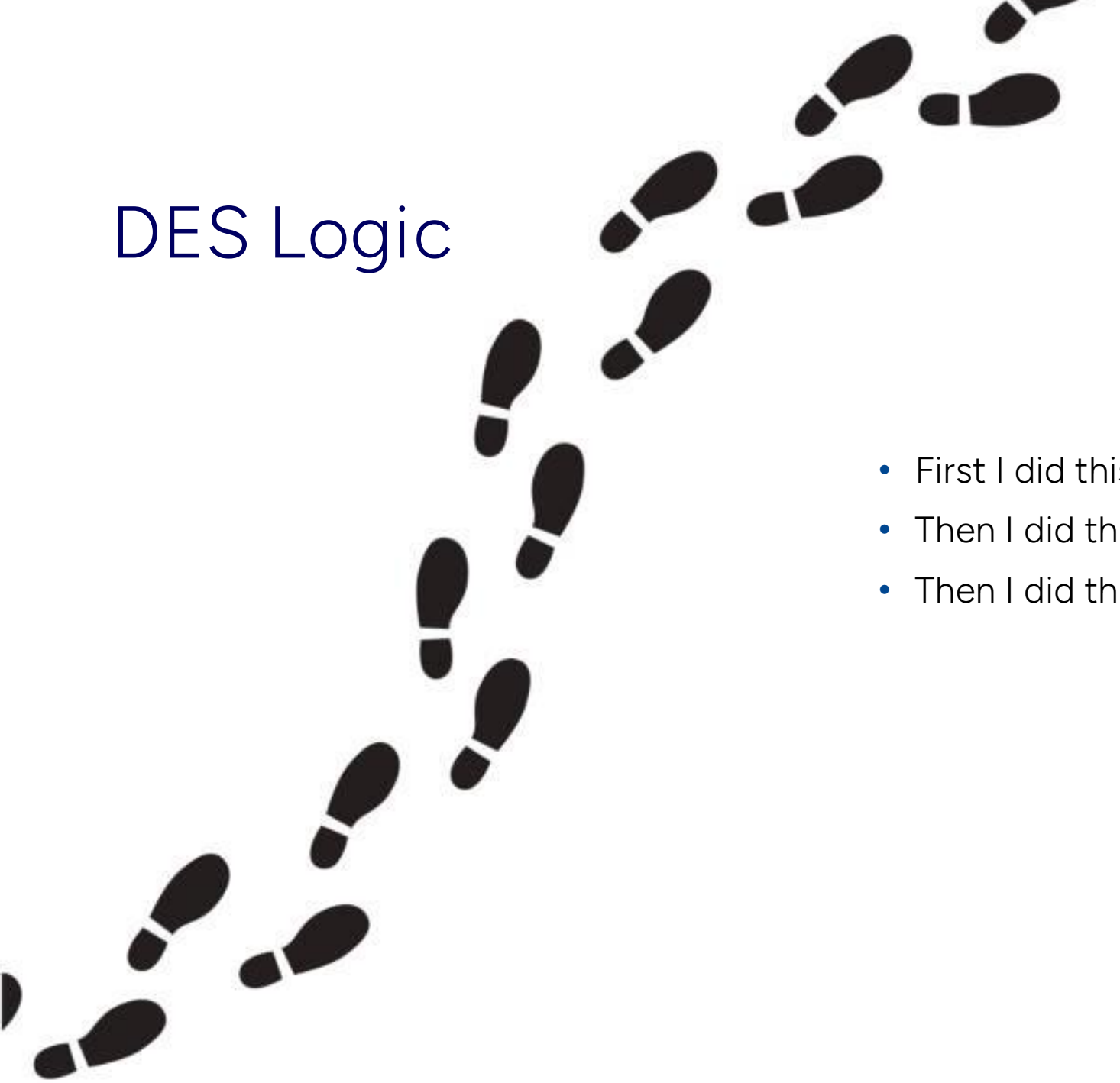


Discrete Dynamic System
or
Discrete Event System

- Doctor
- Nurse
- Room
- Computer

- Utilisation
- Time
- Cost

DES Logic



- First I did this
- Then I did this
- Then I did this



2min

8-9 1m
9-12 2m
12-13 0.5m

1min

ARRIVE

QUEUE

SEIZE BASKET

1per/min

TAKE PRODUCTS

5-60 MIN

SEIZE SCANNER

SELF-CH...

COMPUTER

PAY

RELEASE BASKET

LEAVE

CASH...

CASHIER

PAY

BASKET x100

~~PRODUCTS~~

COMPUTER x10

CASHIER x1

QUEUE → SEIZE CASHIER → RELEASE CATHIER

1-5MIN

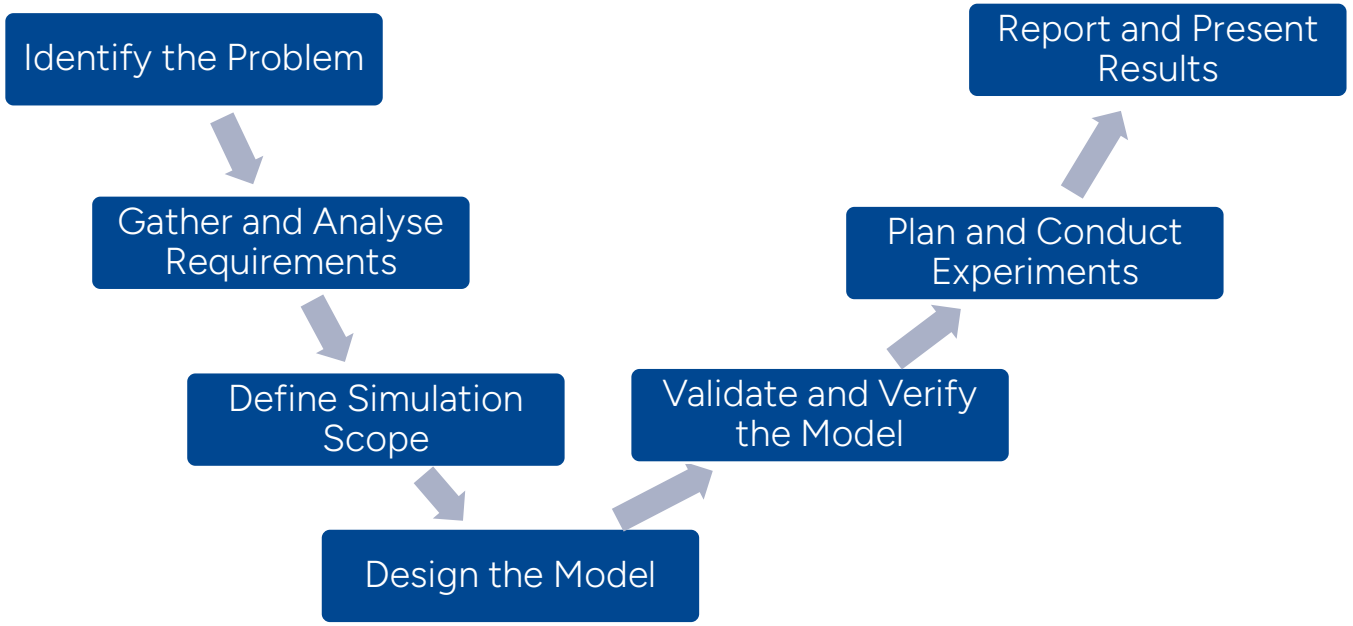
2 + Nr x 0.3 = ___ SEK

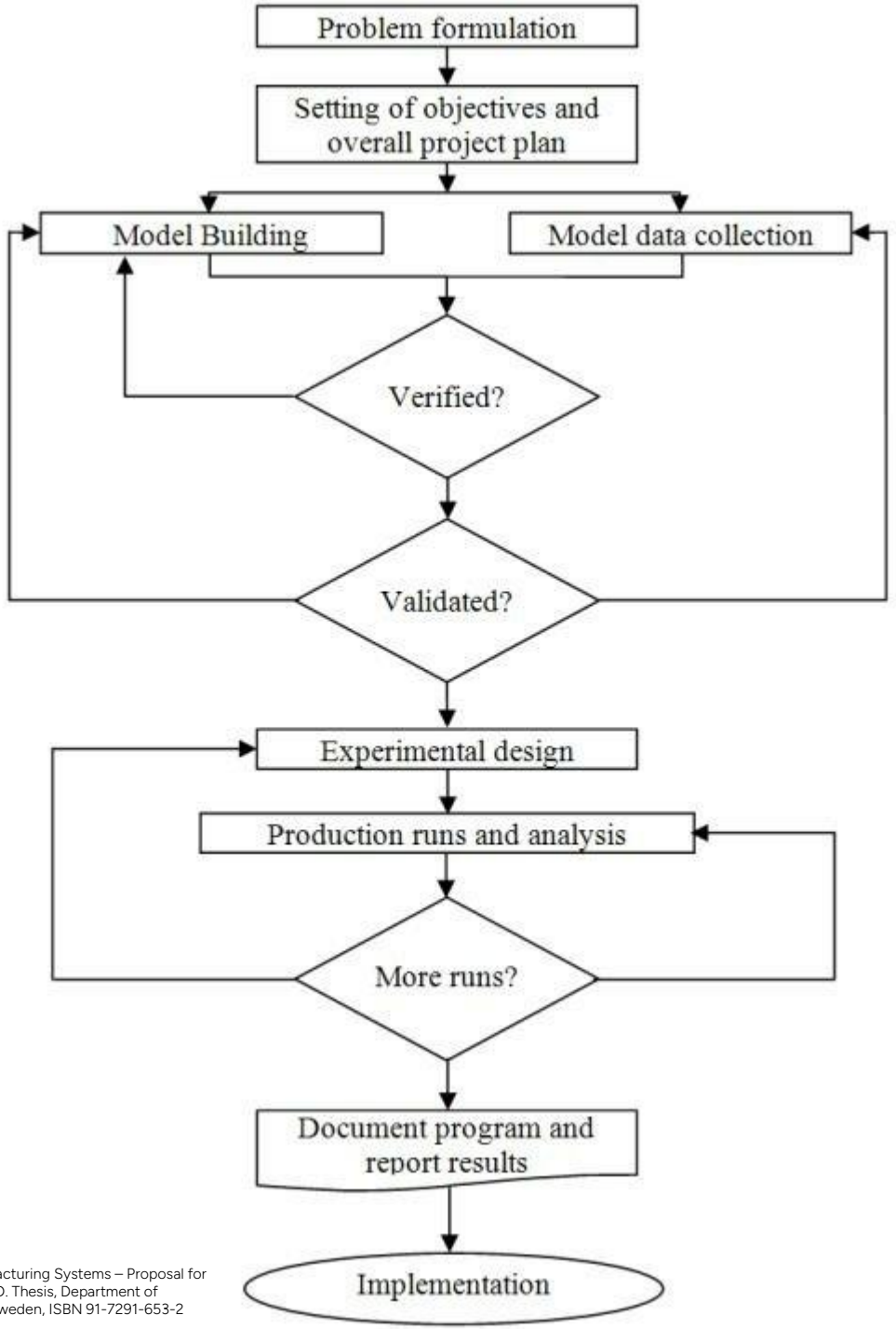
https://youtu.be/b42vD3wdX1c?si=Dz8OxF08yz4DFv57



Current State

How to Simulate





Johansson, B. 2006 On Virtual Development of Manufacturing Systems – Proposal for a Modular Discrete Event Simulation Methodology, PhD. Thesis, Department of Materials and Manufacturing Engineering, Göteborg, Sweden, ISBN 91-7291-653-2

Flowchart - 1: Identify the Problem

- Define Problem Statement
  - Clearly state the issue or scenario to be simulated

- Define Goals and Objectives
  - What outcomes or insights do you want to achieve?

Flowchart - 2: Gather and Analyse Requirements

- Stakeholder Analysis
  - Identify key stakeholders (clinicians, healthcare administrators, IT, etc.)
- Data Collection
  - Gather relevant healthcare data (e.g., patient data, system workflows)
- Identify Constraints
  - Determine resource, time, and technological constraints

Flowchart - 3: Define Simulation Scope

- Determine Simulation Boundaries
  - Define the simulation’s scope (e.g., specific departments, patient types)

- Set Assumptions and Limitations
  - Clarify assumptions and identify any limitations

Flowchart - 4: Design the Model

- Conceptual Model
  - Map out healthcare workflows and decision points
  - Define critical variables, parameters, and potential events
  - Choose between discrete-event, agent-based, or continuous simulation

- Simulation Model
  - Build entities, events, and relationships within the model
  - Integrate real-world data for accurate simulation

Flowchart - 5: Validate and Verify the Model

- Conduct Model Verification
  - Check the model for errors in logic, structure, and consistency
- Validate Against Real Data
  - Ensure the model aligns with real healthcare processes and data
- Consult with Experts
  - Get feedback from stakeholders to confirm accuracy
- Test the Model

Flowchart - 6: Plan and Conduct Experiments

- Define Experimental Scenarios
  - Decide on different scenarios to simulate (e.g., high patient volume, resource shortages)
- Set Key Performance Metrics
  - Choose metrics like wait times, resource utilization, or cost savings
- Run Experiments
  - Perform multiple simulation runs for each scenario
- Analyse Results and Interpret Findings

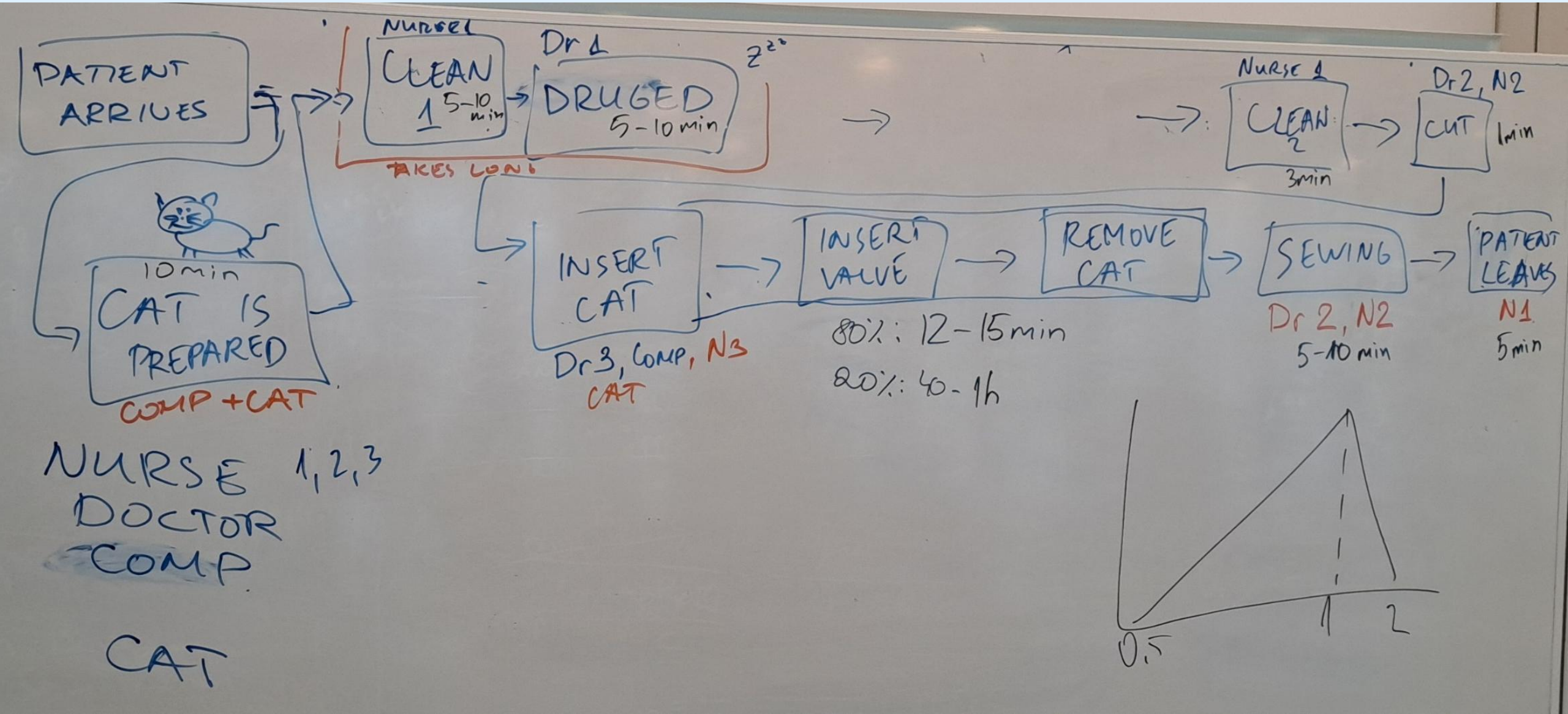
Flowchart - 7: Report and Present Results

- Analyse Data
  - Summarize key insights from each simulation scenario
  - Determine how changes affect healthcare outcomes
  - Create recommendations based on findings

- Prepare Final Report
  - Document findings, methodologies, and conclusions

CAT lab procedure




PATIENT ARRIVES
NURSE 1
CLEAN 1 5-10 min
Dr 1
DRUGED 5-10 min
TAKES LONG
NURSE 1
CLEAN 2 3min
Dr 2, N2
CUT 1min
10min
CAT IS PREPARED
COMP + CAT
INSERT CAT
Dr 3, COMP, N3
CAT
INSERT VALVE
80%: 12-15min
20%: 40-1h
REMOVE CAT
SEWING
Dr 2, N2
5-10 min
PATIENT LEAVES
N1
5min
NURSE 1,2,3
DOCTOR
COMP
CAT
0.5
1
2



Loaders 3

Transferers 7

Controllers 3

Acceptors 3

Orders

Interarrival time: 1 - 5.0 min.

Orders queue capacity: 20